Assignment 1

Please work on the following questions. Write your answers and paste your charts in a separate word file. (It is preferable that you generate the charts in tableau)

1. (50 pts) **Pet Food Sales by State.**
   Pet Fare, Inc. produces high-end refrigerated pet food and ships it anywhere in the United States. Total Pet Fare sales of cat food and dog food by state for the previous year are given in separate worksheets in the file ***Cats&Dogs from Chapter 4 in dataset.zip***. LO 5
   a. (10 pts) Create a geographic information system (GIS) map that uses a sequential color scheme to represent total Pet Fare **cat** food sales by state. Add **State and data** labels. Use **orange** for the color for this **choropleth map**. What information does this chart convey to the audience? (i.e. Which states have relatively high/low cat food sales? is there any geographical patterns?)
   b. (10 pts) Create a geographic information system (GIS) map that uses a sequential color scheme to represent total Pet Fare **dog** food sales by state. Add **State and data** labels. Use **blue** for the color for this **choropleth map**. What information does this chart convey to the audience?
   c. (10 pts) Create a **scatter chart** that enables the audience to better understand the relationship between total **cat** food sales and total **dog** food sales for the states. What does this chart tell you about the relationship between total cat food sales and total dog food sales?
   d. (10 pts) Use color to **highlight Wyoming** in the scatter chart from **part c**. (see this link) What does this chart tell you about total cat food sales and total dog food sales in Wyoming relative to the other states?
   e. (10 pts) What inherent weaknesses do you see in the analysis performed in parts a–c? How would you address these issues? Please draw an improved chart to address these issues. (Hint: Consider the disparities in the sizes of the statewide markets.)

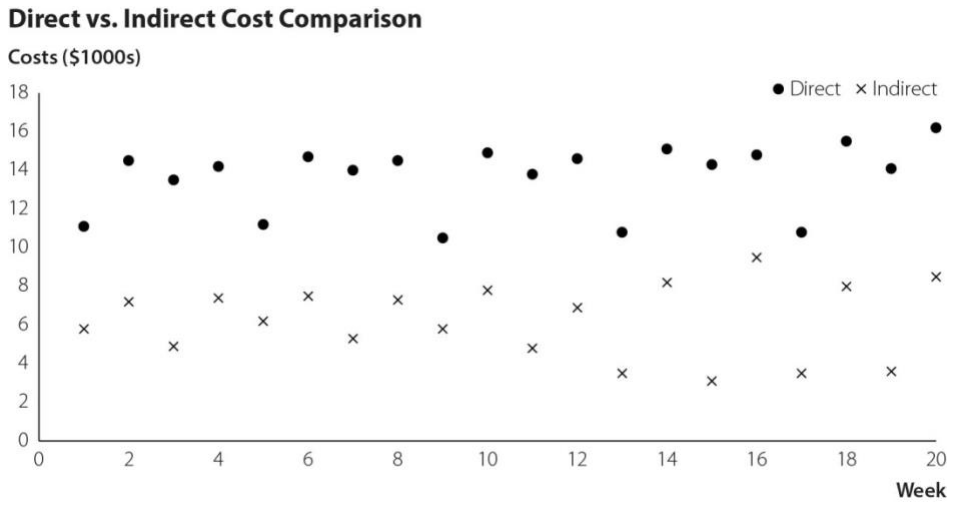
2. (25 pts) **Gross Domestic Product Values over Time**. **(Chapter 3)**
   The following table shows an example of gross domestic product values for five countries over six years in equivalent U.S. dollars ($). LO 3

	Gross Domestic Product (US $)					
Country	**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Year 5**	**Year 6**
Albania	7385937423	8105580293	9650128750	11592303225	10781921975	10569204154
Argentina	169725491092	198012474920	241037555661	301259040110	285070994754	339604450702
Australia	704453444387	758320889024	916931817944	982991358955	934168969952	1178776680167
Austria	272865358404	290682488352	336840690493	375777347214	344514388622	341440991770
Belgium	335571307765	355372712266	408482592257	451663134614	421433351959	416534140346

   a. (10 pts) How could you improve the readability of this table?
   b. (15 pts) The Excel file ***GDPYears from Chapter 3 in dataset.zip*** contains sample data from Year 1 to Year 6 in us $. Create a table that provides all these data for an audience. Format the table to make it as easy to read as possible. *Hint:* It is generally not

important for the audience to know GDP to an exact dollar figure. It is typical to present GDP values in millions or billions of dollars.

3. (25 pts) **Comparing Direct and Indirect Costs at Sackenheim Compressors**.
Sackenheim Compressors, Inc. manufactures air compressors for industrial use. The company is performing an analysis of its direct and indirect costs related to manufacturing a particular type of air compressor. It has gathered cost data for the previous 20 weeks, and it would like to identify any trends in the data. A scatter chart created using the data follows. LO 2, 4



   a. (10 pts) Which Gestalt principle should be used here to make it easier for the audience to identify trends in the data?
   b. (15 pts) Using the chart ***SackenheimChart from Chapter 3 in dataset.zip*** a starting point, create an improved version of this chart by applying the Gestalt principle from part a that makes any trends in the data more obvious to the audience.

General Submission Instructions:
All work must be your own. Copying other people's work or from the Internet is a form of plagiarism and will be prosecuted as such.

You may submit a Microsoft Word (.doc/.docx) document as an attachment using the Canvas Assignment tool, or you may copy and paste your answer into the provided box within the Assignment tool. If you attach a document for your assignment, be sure to include your name in the text of the document and in the name of the document.

- You can only submit once, so make sure you are completely finished before submitting and that you attach the correct word .doc/.docx file.
- Submissions sent by email will NOT be accepted.

Due dates are listed in the Assignment Schedule document.